

# Jimmy Huang

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## Welcome to RenderCV

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Each section title is arbitrary.

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Markdown syntax is supported everywhere. This is **bold**, *italic*, and [link](#).

## Education

|           |                                                                     |                       |
|-----------|---------------------------------------------------------------------|-----------------------|
| <b>MS</b> | <b>Chung Yuan Christian University</b> , Applied Mathematics        | Sept 2020 – June 2022 |
| <b>BS</b> | <b>Chang Jung Christian University</b> , Accounting and Information | Sept 2015 – June 2019 |

## Experience

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| <b>Fortune Fantasy Global Tech .Ltd</b> , Backend Engineer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | On-site                          |
| <ul style="list-style-type: none"> <li>Develop various modern slot games</li> <li>Administrator backend management</li> <li>Explore OpenClaw usage with business</li> </ul>                                                                                                                                                                                                                                                                                                                                                                     | June 2025 – present<br>11 months |
| <b>Fontech</b> , Senior Backend Engineer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Remote                           |
| A real-time, high-throughput, multi-tenant SaaS alarm management service used by various U.S. corporations.                                                                                                                                                                                                                                                                                                                                                                                                                                     | Nov 2024 – May 2025<br>7 months  |
| <ul style="list-style-type: none"> <li>Built from the ground up</li> <li>Spearheaded a team of 3 backend engineers, implementing robust library with clean architecture to improve team's 50% delivery velocity</li> <li>Architected critical system components focusing on horizontal scalability</li> <li>Integrated various AWS infrastructure (Cognito, EC2, KMS, RDS...etc) to reduce operational overhead</li> <li>Developed advanced SQL query optimizations that reduced complex query times from 60s to 0.1s</li> </ul>                |                                  |
| <b>AMI</b> , BMC Firmware Engineer II                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Oct 2024 – Mar 2025<br>6 months  |
| <ul style="list-style-type: none"> <li>Independently operates in BMC</li> <li>Proficient in D-Bus usage</li> <li>Experienced and skilled in developing with Redfish APIs</li> <li>Implemented optimization library for REP architecture, saving up to 50% time and code to do parse and validate</li> <li>Resolved multiple defects across different level modules</li> <li>Proactively proposes solutions to work-related issues and shares them with teams</li> <li>Collaborate with India and US teams with English communication</li> </ul> |                                  |
| <b>Taihe Technology Co., Ltd.</b> , Backend Engineer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Hybrid                           |
| <ul style="list-style-type: none"> <li>Implemented new features and maintained existing microservice projects using PHP and Go</li> <li>Developed new payment and management system using GoFrame and gRPC</li> <li>Quickly familiarized with customized ORM, project architecture, and memSQL</li> <li>Assisted juniors in resolving backend and frontend issues</li> </ul>                                                                                                                                                                    | Mar 2024 – Sept 2024<br>7 months |

**CleNET Technologies**, Software Engineer

- Responsible for testing Pixel cameras, also developing scripts (Python, Shell) to increase automation test coverage from 85% to 95%
- Analyzed test data for potential root causes
- Maintaining the automated testing pipeline

On-site at Google's Camera  
System Team  
Sept 2023 – Mar 2024  
7 months

**Ubitus**, Backend Engineer

- Development of programs based on microservices and cloud infrastructure (Kubernetes, AWS) using Go, Python, and Node.js.
- Deployment and scripting for setting up cloud services and various services using Docker, Ansible, and Helm.
- Development and maintenance of an existing in-game content management system using Node.js, Go, and Python.
- Establishing a real-time Twitter comment monitoring system for internal use using Python and Prometheus.
- Developing AI-related services and applications like stable-diffusion using Python.
- Built up log collectors and workflow mechanisms using Fluentd and Airflow.
- Writing unit tests, integration tests, and end-to-end tests using K6.

Taiwan  
June 2021 – Aug 2023  
2 years 3 months

**Chung Yuan Christian University**, Teaching Assistant

- Object Oriented Programming TA
- Introduction to Data Science TA

Taiwan  
Sept 2021 – June 2022  
10 months

## Projects

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**Alarm system**

The project built on well-known camera corp for alarm management.

- Built the backend infrastructure and db schema to fit business logic.
- Optimized the index creation to ensure low RTs, reducing them from 50s to 3s.
- Built a real-time alarm monitoring API using Go (Gin) and GORM, supporting complex alarm lifecycles and multi-tenant data isolation.
- Integrated AWS Cognito and custom JWT middleware to manage secure user authentication and role-based access control.
- Automated schema migrations and versioning across MySQL using Atlas and GORM.
- Architected a secure Webhook and API Key system to enable third-party integrations and automated reporting.
- Standardized API documentation with Swagger (swag) and implemented structured logging using Zerolog for rapid debugging.
- Containerized the application with Docker and automated deployment workflows using Makefiles and Shell scripts.

Associated with Fontech  
Nov 2024 – present

**AuthZ**

- Architected a real-time, Zanzibar-inspired ReBAC engine supporting complex, recursive permission schemas (RBAC/ABAC), achieving 10k+ RPS per service with sub-1ms median latency.
- Engineered a memory-first, sharded graph architecture that eliminated lock contention and facilitated efficient tree-based relationship exploration for authorization tuple operations.
- Optimized database performance by implementing Redis-backed Cuckoo Filters for probabilistic existence checks, preventing redundant writes and slashing expensive duplicate lookups.
- Built an event-driven state synchronization pipeline utilizing Kafka and Debezium (CDC) to stream tuple updates, ensuring real-time consistency across a distributed cluster.
- Integrated comprehensive observability with OpenTelemetry (OTel) for distributed tracing and metrics, streamlining troubleshooting in high-throughput production environments.

July 2025 – Aug 2025

## RAG

July 2025 – Aug 2025

This project implements a RAG (Retrieval-Augmented Generation) system to accelerate software development tasks like architecture comprehension and debugging. It leverages a large language model (LLM) to understand and analyze a codebase.

- Used LangChain to generate vector embeddings from the source code.
- Stored embeddings in a PostgreSQL database using the PGVector extension for efficient similarity searches.
- A self-hosted Llama model serves as the core LLM for generation and reasoning.
- Retrieves relevant code snippets from the vector store and provides them as context to the LLM upon query.

## The application of tree-based model for well interpretation strategy

Associated with Chung Yuan

Christian University

Jan 2022 – June 2023

- Engineered an automated data extraction pipeline using a web crawler to scrape Grandmaster-level replays, processing game screens every two seconds to create a continuous dataset of 3,309 matches.
- Developed a computer vision module utilizing Hough Circles and custom RGB frame filters to detect and extract dynamic in-game features.
- Scaled the image dataset for model training by applying extensive data augmentation techniques, including background splitting and randomization, to generate over 52,500 training samples.
- Trained and evaluated multiple machine learning models—including CNNs, XGBoost, Random Forest, and GradientBoost—using 8-fold cross-validation to classify game states.
- Designed an interpretable decision-tree-based recommendation engine that analyzed real-time feature importance, successfully identifying strategies that increased simulated win rates by 18% to 34%.

## Vistrace

- Engineered a high-performance ARPG engine in Rust/Bevy with strict DOD and 100% ECS compliance, optimizing the WASM pipeline for sub-second web deployment.
- Architected an AI-driven Agentic Workflow utilizing 10+ specialized AI sub-agents to autonomously execute TDD logic, verify builds, and manage cross-module dependencies, reducing development cycles by 30%.
- Developed 12+ custom Rust CLI tools and static analysis guardrails that eradicated circular dependencies and automated 80% of the game's content ingestion pipeline.
- Automated end-to-end testing and documentation by combining AI scenario generation with Playwright/Rust, achieving 95%+ test coverage on core systems, cutting manual QA time by 60%, and maintaining 100% automated documentation accuracy.
- Integrated generative AI pipelines to accelerate asset creation, deploying 30+ production-ready UI and background assets and reducing deployment time from days to minutes.

## Skills

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**Languages:** Python, C++, CUDA, Rust, Julia

**ML Frameworks:** PyTorch, JAX, TensorFlow, Triton, ONNX

**Infrastructure:** Kubernetes, Ray, distributed training, AWS, GCP

**Research Areas:** Neural architecture search, model compression, efficient inference, multi-agent RL